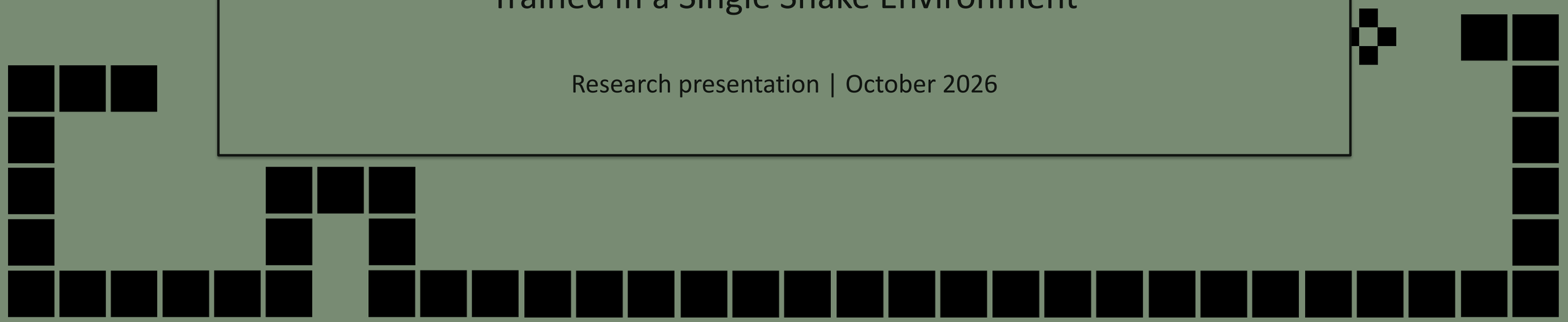
SNAKE

CROSS-LEVEL GENERALIZATION

Evaluating a Dueling Double DQN

Trained in a Single Snake Environment

Research presentation | October 2026



THE RESEARCH QUESTION

Can one trained policy handle different Snake levels?

RQ1 / SOURCE

How well does the agent perform on Level 4?

RQ2 / TRANSFER

Does the frozen policy reach targets on five unseen levels?

RQ3 / RELIABILITY

How do scores and failure modes vary across layouts?

Source: project implementation and latex/main.tex

STUDY AT A GLANCE

TRAIN

Level 4 only

One training seed

Dueling Double DQN

FREEZE

27 state features

3 relative actions

Greedy evaluation

EVALUATE

6 levels × 100 episodes

Best + final + random

1,800 episodes total

Single source environment → frozen policy → unseen-level evaluation

Source: project implementation and latex/main.tex

SIX ENVIRONMENTS, ONE SOURCE

Level	Board	Obstacles	Target	Role
L1	15 × 15	0	25	Unseen
L2	20 × 20	8	35	Unseen
L3	24 × 24	16	50	Unseen
L4	30 × 30	20	None	Training + evaluation
L5	32 × 32	34	100	Unseen
L6	36 × 36	68	150	Unseen

Board size, obstacle geometry and target change together.

Source: project implementation and latex/main.tex

HOW THE AGENT SEES THE GAME

Feature group	Count	Purpose
Danger + ray clearance	6	Local collision awareness
Heading	4	Current movement direction
Food direction + displacement	6	Navigation toward food
Head position + body length	3	Normalized geometry
Reachable safe areas	3	Space after candidate actions
Tail direction + distance	5	Body / tail awareness

Source: project implementation and latex/main.tex

RELATIVE ACTIONS AND REWARDS

STRAIGHT

Continue in the current direction.

TURN RIGHT

Rotate relative to the current heading.

TURN LEFT

Rotate relative to the current heading.

Food +10 **Collision −20** **Closer +0.015** **Otherwise −0.025**

No-food timeout: 2 × board cells; counter resets after eating.

Source: project implementation and latex/main.tex

DUELING DOUBLE DQN

ENCODE

27 features

→ 256 ReLU

→ 256 ReLU

DUELING

State value V(s)

Action advantages A(s,a)

→ three Q-values

DOUBLE DQN

Online network selects

Target network evaluates

→ Huber loss update

Q(s,a) = V(s) + A(s,a) − mean A(s,·)

Sources: van Hasselt et al. (AAAI 2016); Wang et al. (ICML 2016); project agent.

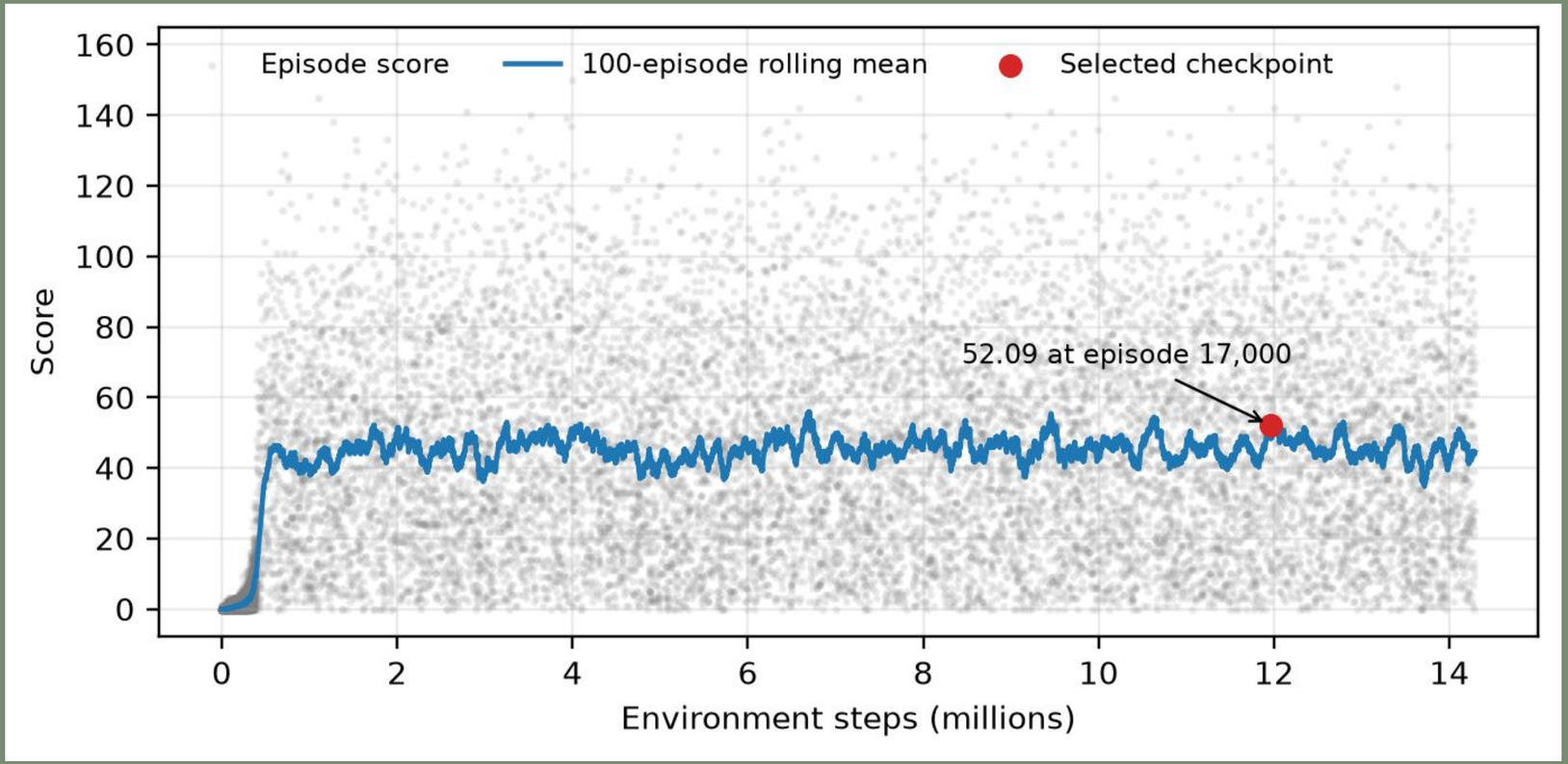
TRAINING CONFIGURATION

Setting	Value	Setting	Value
Learning rate	0.0003	Discount	0.995
Batch size	256	Replay capacity	300,000
Warm-up steps	5,000	Target update	5,000 steps
Update frequency	Every 2 steps	Exploration ε	1.00 → 0.01
Environment steps	14,293,725	Training seed	0

Selected checkpoint: episode 17,000 | training rolling mean: 52.09

Source: project implementation and latex/main.tex

LEARNING ON LEVEL 4



52.09

Selected checkpoint

100-episode training mean

Training statistics are separate from evaluation results.

Source: project implementation and latex/main.tex

FROZEN-POLICY EVALUATION

- 100 episodes per policy–level pair; seeds 1234–1333.
- Best DQN, final DQN and uniform random policy.
- Frozen weights, greedy actions and headless evaluation.
- Scores, target success, normalized scores and failure causes.
- 95% Wilson intervals for target success rates.

Source: project implementation and latex/main.tex

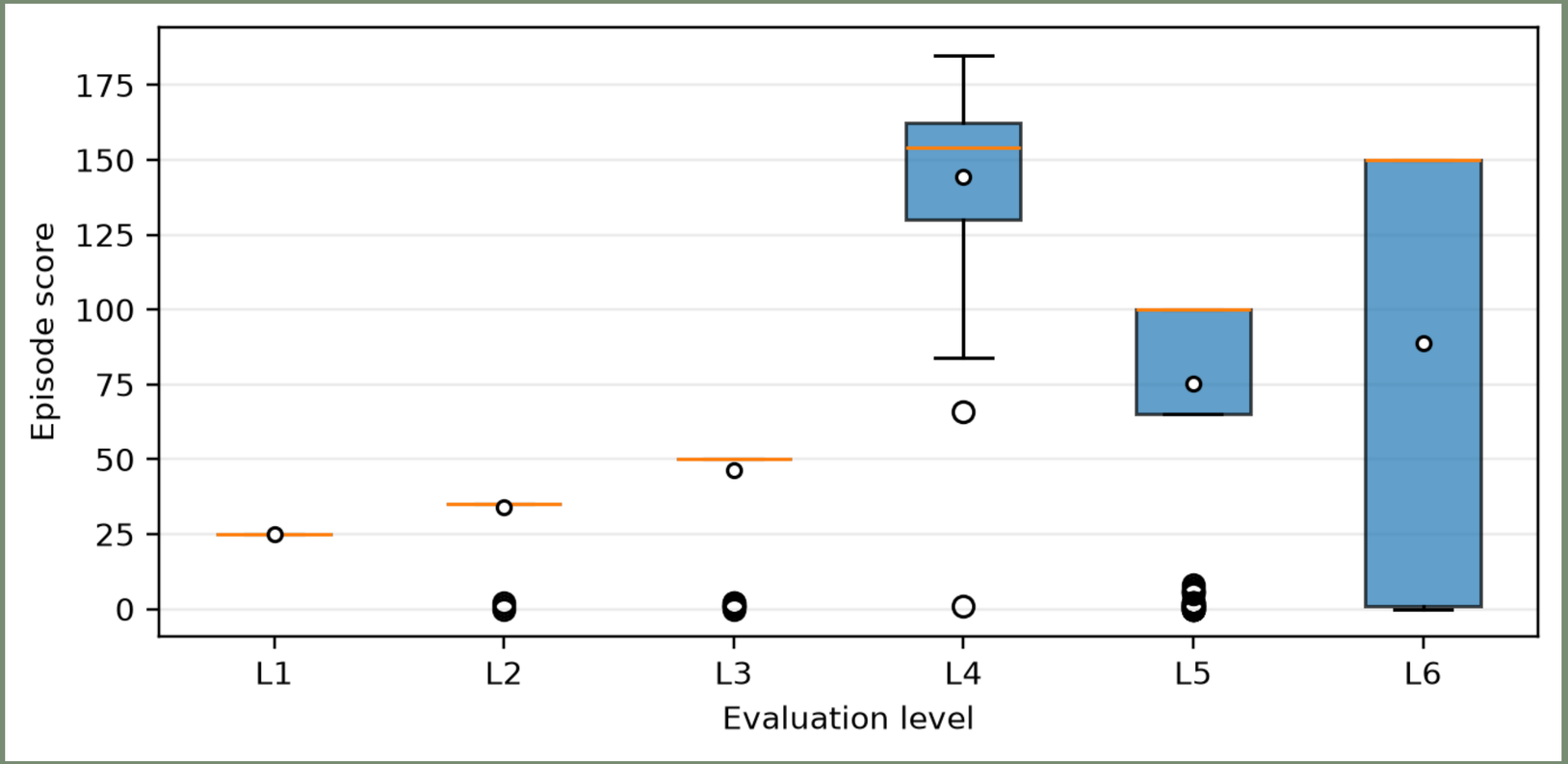
ZERO-SHOT SUCCESS ACROSS LEVELS



Strong transfer on smaller levels; less reliable on L5 and L6.

Source: results/cross_level_evaluation/summary.csv | best_dqn | n = 100 per level

SCORE DISTRIBUTIONS REVEAL INSTABILITY



LEVEL 6

Median: 150

P25: 1

P75: 150

L4 mean ± SD: 144.49 ± 27.28 | no target cap

Source: project implementation and latex/main.tex

CHECKPOINT CHOICE CHANGES TRANSFER

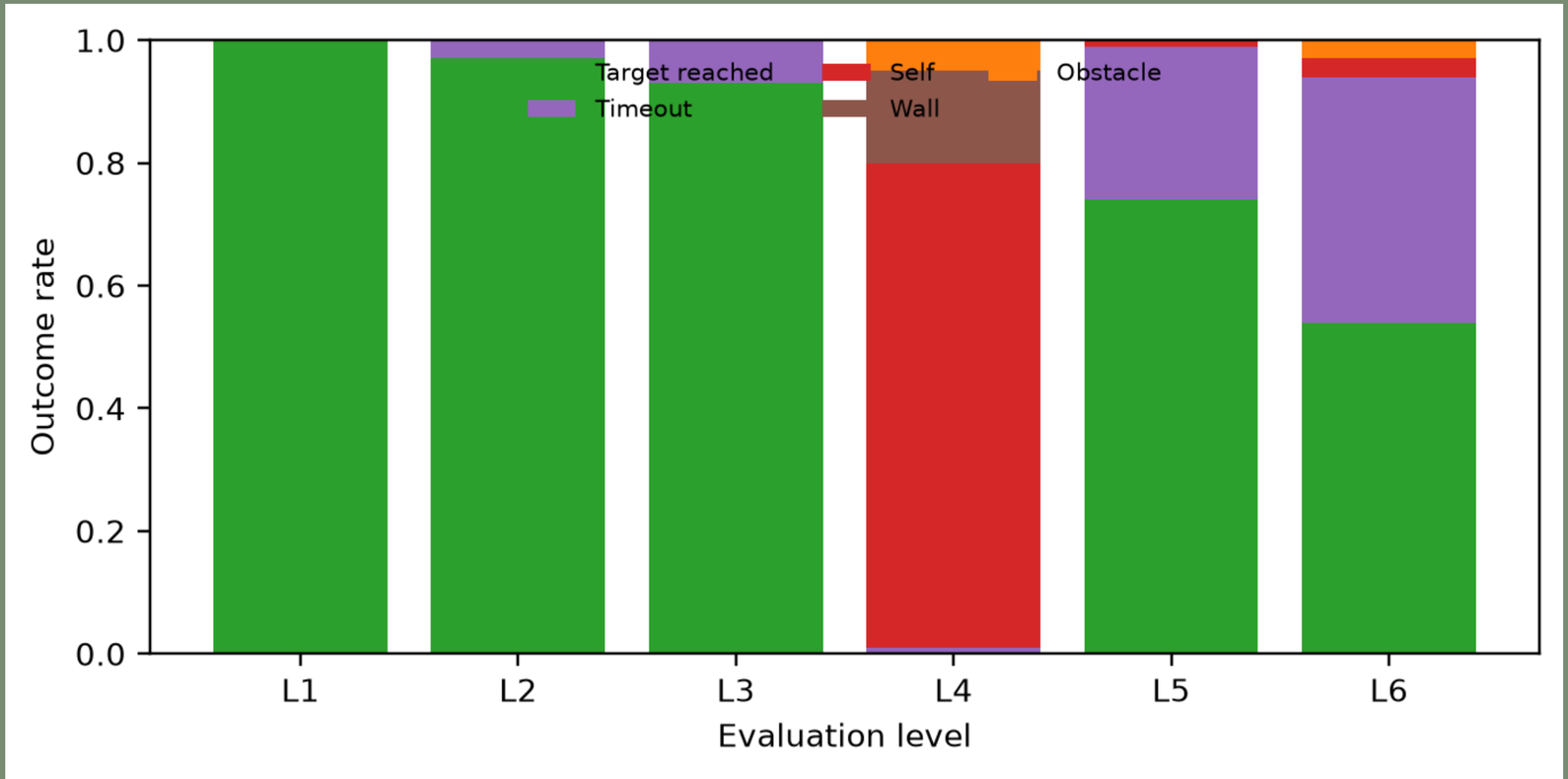
Metric	Selected DQN	Final DQN
L4 mean score	144.49	153.83
L2 success	97%	99%
L3 success	93%	83%
L5 success	74%	65%
L6 success	54%	52%

Better source-level score did not consistently improve transfer.

Random baseline: mean scores 0.02–0.10; no target reached.

Source: project implementation and latex/main.tex

WHERE THE POLICY FAILS



TIMEOUT

L5: 25%

L6: 40%

L4 self-collision: 79%

L4 has no target cap; its termination profile is different.

Source: project implementation and latex/main.tex

WHAT THE EVIDENCE SUPPORTS

RQ1

L4: 144.49 ± 27.28

Median score: 154

Occasional early failure

RQ2

100%, 97%, 93% on L1–L3

74%, 54% on L5–L6

Transfer is incomplete

RQ3

Timeouts rise on L5–L6

Level factors co-vary

Causal effects unresolved

A shared representation enables execution; evaluation establishes performance.

Source: project implementation and latex/main.tex

LIMITATIONS AND NEXT EXPERIMENTS

Current limitation	Next experiment
One training seed	Repeat independent training runs
One source level; five unseen levels	Evaluate additional held-out layouts
Board, obstacles and target co-vary	Change one factor at a time
No representation / reward ablation	Remove feature groups and shaping
Training-score checkpoint selection	Use a separate validation protocol

Then compare curricula / domain randomization under equal step budgets.

Source: project implementation and latex/main.tex

CONCLUSION

- One Level-4-trained network transferred without fine-tuning.
- Success: 100%, 97%, 93%, 74%, 54% on L1, L2, L3, L5, L6.
- Larger unseen levels exposed high variance and timeouts.
- More training did not consistently improve cross-level success.

Substantial transfer within this suite — incomplete robustness.

Source: project implementation and latex/main.tex

SELECTED REFERENCES

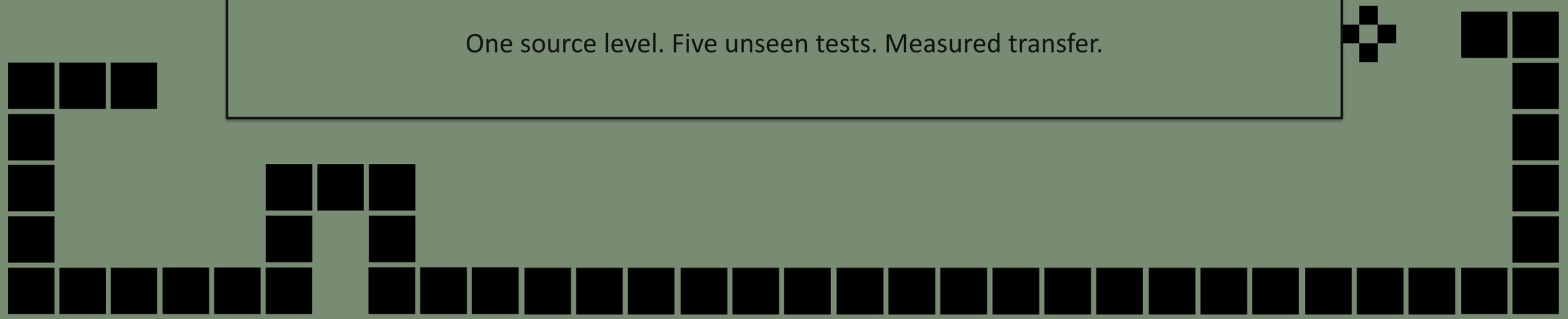
- Mnih et al. (2015). Human-Level Control through Deep Reinforcement Learning. Nature.
- van Hasselt et al. (2016). Deep Reinforcement Learning with Double Q-Learning. AAAI.
- Wang et al. (2016). Dueling Network Architectures for Deep Reinforcement Learning. ICML.
- Cobbe et al. (2019). Quantifying Generalization in Reinforcement Learning. ICML.
- Henderson et al. (2018). Deep Reinforcement Learning That Matters. AAAI.

Full bibliographic entries and links: latex/references.bib

THANK YOU

QUESTIONS?

One source level. Five unseen tests. Measured transfer.



BACKUP / DETAILED PRIMARY RESULTS

Level	Mean ± SD	Median	Success	Norm. score
L1	25.00 ± 0.00	25.0	100%	1.000
L2	33.98 ± 5.80	35.0	97%	0.971
L3	46.58 ± 12.47	50.0	93%	0.932
L4	144.49 ± 27.28	154.0	N/A	N/A
L5	75.40 ± 42.29	100.0	74%	0.754
L6	88.71 ± 71.77	150.0	54%	0.591

Source: project implementation and latex/main.tex

BACKUP / ANTICIPATED QUESTIONS

Question	Evidence-based answer
Why call these levels unseen?	Only L4 supplied training transitions in this run.
Does 100% mean guaranteed success?	No; 100/100 gives a 95% Wilson interval of 96.3–100%.
Is the method better than all baselines?	Only random and final-checkpoint comparisons were evaluated.
Do features cause generalization?	Plausible, but no ablation isolates their effect.
Why no PPO comparison?	The research scope is single-source DQN transfer.

Source: project implementation and latex/main.tex